

Technology News

Universal robotic gripper

The human hand is an amazing machine that can pick up, move and place objects easily, but for a robot, this 'gripping' mechanism is a vexing challenge. Opting for simple elegance, researchers from Cornell University, the University of Chicago and iRobot Corp. have created a versatile gripper using everyday ground coffee and a latex party balloon, bypassing traditional designs based on the human hand and fingers. They call it a universal gripper, as it conforms to the object it's grabbing, rather than being designed for particular objects.

The universality of the gripper makes future applications seemingly limitless, from the military using it to dismantle explosive devices or to

move potentially dangerous objects, robotic arms in factories, on the feet of a robot that could walk on walls or on prosthetic limbs.

Here's how it works: An everyday party balloon filled with ground coffee—any variety will do—is attached to a robotic arm. The coffee-filled balloon presses down and deforms around the desired object, and then a vacuum sucks the air out of the balloon, solidifying its grip. When the vacuum is released, the balloon becomes soft again, and the gripper lets go.

LED light affects human health

Recent researches carried out by three organisations including Illumination

Engineering Society (IES), French Agency for Food, Environmental and Occupational Health & Safety (ANSES) and the US Department of Energy have revealed that the blue light component in white LEDs causes toxic stress to the retina.

Blue light has been found to cause a photochemical risk to the eye, the level of which depends on the accumulated dose of blue light to which the person has been exposed—generally the result of low-intensity exposure repeated over long periods. In particular, three groups are at risk: children, populations which are already light-sensitive and workers likely to be exposed to high-intensity lighting.

The other main risk is from glare. For indoor lighting, it is generally agreed that luminance higher than 10,000 cd/m² causes visual discomfort whatever be the position of the lighting unit in the field of vision. Because the emission surfaces of LEDs are highly-concentrated point sources, the luminance of each individual source can be a thousand times higher than the discomfort level. The level of direct radiation from this type of source can therefore easily exceed the level of visual discomfort.

Laptops may affect male fertility

If a recent study is anything to go by, men who regularly put their laptops on their laps when working may face reproductive problems later in their life.

"Millions and millions of men are using laptops now, especially those in the reproductive age range. Within ten or 15 minutes, their scrotal temperature is already above what we consider safe, but they don't feel it," said Yelmin Sheynkin, a urologist at the State University of New York at Stony Brook who led the study published in



Universal robotic gripper (Photo credit: John Amend, Cornell University)